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Impact of negative word of mouth on consumers' attitude. Moderating role of advertising under cognitive involvement conditions

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ABSTRACT

The proliferation of internet technology and social media platforms has revolutionized consumer-brand interactions, enabling widespread participation in online brand conversations and significantly amplifying the impact of word-of-mouth communication. Therefore, they actively post and increasingly rely on online product reviews, particularly when the reviews are negative. These product reviews carry cognitive and affective information that can be found in comparative and non-comparative formats. In this study, we focused on cognitive information available online in comparative and non-comparative formats. This study examines the impact of negative word-of-mouth communication on consumer attitudes and the moderating role of attribute-based advertising under comparative and non-comparative cognitive involvement conditions. A two cognitive negative word-of-mouth and two cognitive attribute-based advertisement mixed designs were used to empirically test the proposed research model. Data were analyzed using a two-way ANOVA in SPSS to examine the interaction effects between NWOM and advertising formats on consumer attitudes. The findings reveal that attribute-based advertising communication moderates the negative impact of NWOM communication on consumer attitudes under comparative and non-comparative cognitive involvement conditions. The theoretical implications, practical implications, and recommendations are discussed for further research in the field of online media communication.

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Introduction

With technological advancement in today's marketing world, electronic word of mouth (eWOM) is emerging as a persuasive marketing tool. Electronic word-of-mouth (eWOM) refers to any positive or negative statement made by consumers about a product or company, shared electronically through digital platforms such as social media, online forums, or review websites (Ma et al., 2022; Ribeiro & Kalro, 2023). The term eWOM also defines as any positive or negative statement made by potential, actual, or former customers about a product or company (Ma et al., 2022; Özsomer et al., 2024). It is a form of consumer-to-consumer communication that is visible, scalable, and persistent online and can greatly influence other consumers' perceptions and decisions (Bickart & Schindler, 2001; Ngo et al., 2024; Wang & Sun, 2010; Wiese et al., 2020).

Consumers' desire for social interaction motivates them to spread eWOM (Amul, 2020) so it is not in the control of marketers to hold back the negative word of mouth (NWOM), which is a specific subset of eWOM containing unfavorable opinions or dissatisfaction expressed online. Dissatisfaction leads to the generation of NWOM; hence, dissatisfied consumers spread NWOM information more often (Dondolo, 2017; Lovett et al., 2019; Ribeiro & Kalro, 2023). While marketers cannot directly prevent NWOM from

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being shared online, they can manage its impact through strategic communication efforts, such as targeted advertising and content responses (Khattoon et al., 2024).

This study specifically focuses on NWOM that carries cognitive information, that is, rational, factual content, consumers may use to assess and compare product attributes. Consumers tend to seek other consumers' reviews when it comes to evaluating unfamiliar products in online settings; if these reviews are negative, they have the potential to alter their decisions about the product (Ma et al., 2022; Tan & Chen, 2023). Before making any decision, consumers evaluate different products in the same category by comparing their attributes. Therefore, it has eventually become an ultimate challenge for marketers to harness consumers' positive attitudes toward a product by managing NWOM communication online. When left unaddressed, NWOM can spread rapidly and shape public perception negatively, ultimately eroding trust in the brand and damaging its perceived value or equity.

Furthermore, marketers should be careful in spending their budget on advertising to counter the NWOM generated by dissatisfied consumers (Azemi et al., 2020; Khattoon et al., 2024). In the long run, NWOM can propagate as a result of consumer dissatisfaction. Therefore, it can damage the brand's advertising equity, and it should be efficiently encountered by marketers efficiently (Wang et al., 2021). In literature, attribute-based advertising is defined as advertising that emphasizes specific product features and performance characteristics, and is often used as an effective marketing communication tool to reduce the damage caused by NWOM (De Jans et al., 2018; Hussain et al., 2023; Kelly et al., 2021).

Although previous studies have examined the impact of NWOM on consumer attitudes, the interactive effect of NWOM and advertising formats, particularly comparative advertising (Adcc) and non-comparative advertising (Adnc), remains underexplored. This study aims to fill this gap by investigating how different types of NWOM (comparative vs. non-comparative) interact with attribute-based advertising to influence consumer attitudes. Further, the impact of 'NWOM communication' on 'consumer attitude', and the moderating role of 'attribute-based advertising' under comparative and non-comparative cognitive involvement conditions were also examined.

According to the Elaboration Likelihood Model, when individuals have high motivation and ability, they engage in central route processing, leading to deeper cognitive engagement with message content (Petty & Cacioppo, 1986). It is elucidated in the literature that direct comparisons between the brands trigger high cognitive involvement, which may be left unnoticed if non-comparative advertising is applied; hence, comparative advertising (Adcc) is more likely to engage central route processing due to its diagnostic nature (Kowalczyk et al., 2021). Previous research has confirmed that negative word of mouth influences consumers far more effectively than positive word of mouth communication (Ahmadi, 2019; Azemi et al., 2020; Nam et al., 2020). It has also been observed that negative word of mouth is transmitted more rapidly than positive word of mouth (Gilal et al., 2018). Negative electronic word of mouth communication has been suggested to play a major role in shaping consumer attitude toward a product or service (Yu et al., 2019). Previous research studies have already examined general negative statements, such as negative word of mouth communication (Keh & Sun, 2018).

Several researchers have examined the impact of comparative versus non comparative advertising on consumer attitudes, and concluded that the impact is inconsistent (Ismagilova et al., 2020; Moore & Lafreniere, 2020; Yen & Tang, 2019). Attitude is defined as 'learned predispositions to respond to an object or class of objects in a favorable or unfavorable way,' which indicates that the attitude toward an object is formed in response to the information received by an individual, consumer, or receiver (Bright et al., 1993; Hoque et al., 2018). Hence, it can be concluded that attitude formation is a learned behavior, and any advertising, eWOM communication, and marketing messages have the potential to influence it (Robinson, 2018).

The influence of negative word-of-mouth (NWOM) on consumer behavior has been extensively documented in prior research, with studies emphasizing its stronger impact compared to positive word-of-mouth (Amul, 2020; Yu et al., 2019). Although positive word-of-mouth (PWOM) can influence consumer perceptions, prior research indicates that NWOM is more diagnostic and impactful in shaping consumer attitudes (Gilal et al., 2018; Ma et al., 2022). Consumers are more likely to remember and act upon negative reviews than positive ones. Given this stronger influence, the present study focuses solely on NWOM to understand how brands can mitigate its adverse effects through strategic advertising. Similarly, advertising strategies, including comparative and non-comparative formats, have been studied for their

effectiveness in shaping consumer attitudes (Wiese et al., 2020). However, how these distinct advertising formats (Adcc vs. Adnc) interact with NWOM content (comparative vs. non-comparative) under different levels of cognitive involvement remains theoretically and empirically unclear.

While NWOM is recognized as a powerful driver of consumer attitudes, its interplay with attribute-based advertising formats has not been systematically analyzed, leaving a gap in understanding the mitigating potential of advertising in the presence of NWOM. This study specifically focuses on attribute-based advertising as a strategic response to negative word-of-mouth (NWOM). While various strategies such as influencer marketing and public relations can help counter NWOM, advertising provides a controlled and structured approach where brands can proactively frame cognitive information to mitigate the negative effects of NWOM. By examining the interaction between NWOM and advertising formats, this research contributes to understanding how brands can shape consumer attitudes despite exposure to NWOM.

A significant body of literature has focused on the effectiveness of comparative versus non-comparative advertising, with mixed findings suggesting cultural and contextual factors may influence consumer perceptions (Hussain et al., 2023; Khatoon et al., 2024; Yen & Tang, 2019). At the same time, studies have shown that NWOM is perceived as more diagnostic and impactful than positive reviews (Gilal et al., 2018; Tan & Chen, 2023). Despite these insights, there is limited empirical evidence on whether attribute-based comparative or non-comparative advertising can alter consumer attitudes in the face of NWOM, and if so, under what conditions. This research gap is particularly critical in digital environments where NWOM and advertising frequently coexist and simultaneously influence consumer decision-making.

Moreover, the cultural context of Pakistan, where comparative advertising is less prevalent and often deemed unethical, presents a unique opportunity to examine this interaction. Prior research has largely been conducted in Western contexts, where comparative advertising is a common practice (Birim et al., 2024; Huh & Malthouse, 2020; Liu, 2023; Zhou, 2013). The findings from this study contribute to filling this gap by exploring the interactive effects of NWOM and advertising formats in a non-Western context. By examining how consumers process cognitive information from NWOM and advertising in a market where comparative formats are less familiar, this study adds a valuable dimension to the global understanding of advertising effectiveness.

The existing literature has explored the impact of eWOM communication carrying cognitive information on consumer attitudes through the application of the Elaboration Likelihood Model (ELM) (East et al., 2008; Kowalczyk et al., 2021; Liu et al., 2018). According to ELM, messages containing cognitive information are processed through central cues, requiring greater elaboration, which positions receivers as active participants in message processing (Petty and Cacioppo 1986). Additionally, this study incorporates the Accessibility-Diagnosticity Model, which suggests that brand-related information presented in a comparative format is perceived as more diagnostic than non-comparative information (Ahmad, 2019). Consequently, negative information tends to hold greater diagnostic value than positive information, requiring more careful evaluation (Ahmad, 2019; Lynch & Feldman, 1988).

This study addresses the above mentioned gaps by focusing on the interactive effects of NWOM and advertising formats on consumer attitudes under varying conditions of cognitive involvement. The integration of the Elaboration Likelihood Model (Petty & Cacioppo, 1986) and the Accessibility-Diagnosticity Model (Lynch & Feldman, 1988) provides a theoretical framework to explore these dynamics. By doing so, this research not only contributes to the academic discourse on NWOM and advertising but also offers practical insights for marketers aiming to design effective strategies to counter NWOM in digital settings.

Despite extensive research on consumer behavior and advertising, significant gaps remain in understanding the interaction between negative word-of-mouth (NWOM) and attribute-based advertising in comparative and non-comparative formats. While the Elaboration Likelihood Model and Accessibility-Diagnosticity Model provide theoretical frameworks for analyzing consumer information processing, their application to the interactive effects of NWOM and advertising under varying cognitive involvement conditions is limited. Methodologically, there is a lack of quasi-experimental studies that use carefully validated stimuli to explore the dynamics of consumer attitude formation in response to NWOM and advertising. Additionally, practical strategies for marketers to effectively mitigate the adverse effects of NWOM through tailored advertising remain underexplored, particularly in contexts where NWOM significantly influences consumer perceptions. Furthermore, little is known about how these dynamics operate in cultural contexts like Pakistan, where comparative advertising is less prevalent and often viewed

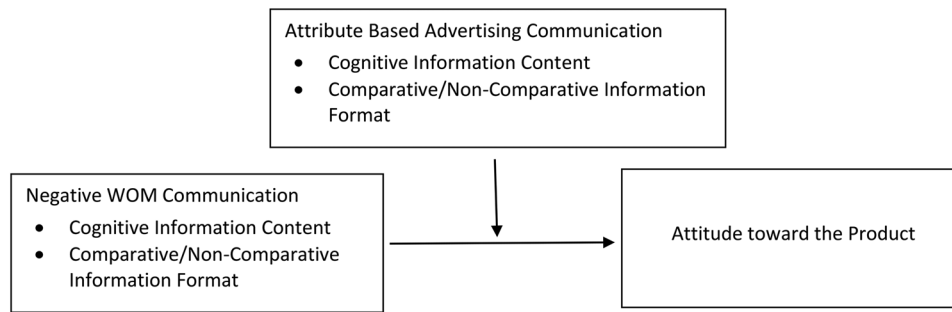


Figure 1. Proposed research model.

as unethical. Addressing these gaps is essential to advancing theoretical understanding and providing actionable insights for practitioners.

Therefore, this research has been conducted to address these gaps. The study employs a rigorous experimental design incorporating four distinct conditions: ‘negative word-of-mouth comparative content and advertising comparative content (NWOMcc and Adcc),’ ‘negative word-of-mouth comparative content and advertising non-comparative content (NWOMcc and Adnc),’ ‘negative word-of-mouth non-comparative content and advertising comparative content (NWOMnc and Adcc),’ and ‘negative word-of-mouth non-comparative content and advertising non-comparative content (NWOMnc and Adnc).’ Based on these conditions, the conceptual framework illustrated in [Figure 1](#) was developed, leading to the formulation of hypotheses H1 and H2, which are tested in this research study.

Proposed research model

The research framework outlines an interactive approach to examining the effects of ‘negative word-of-mouth communication’ on ‘product attitudes,’ while also considering the ‘moderating influence of attribute-based advertising.’ The interaction between NWOM and advertising formats was theorized using the Elaboration Likelihood Model and Accessibility-Diagnosticity Model. NWOMcc was expected to be more damaging due to higher diagnosticity, while Adcc was expected to counter this through equally diagnostic content. Conversely, NWOMnc was considered less influential, and Adnc was expected to reinforce neutrality, leading to more favorable attitudes

Hypothesis 1

NWOMcc is likely to result in a more negative attitude toward the product than NWOMnc.

Hypothesis 2

Attribute-based cognitive advertising moderates the relationship between NWOM and consumer attitudes toward the product. Specifically, individuals exposed to NWOMcc and Adcc are expected to form a more favorable attitude toward the product than those exposed to NWOMcc and Adnc. Conversely, individuals exposed to NWOMnc and Adnc are likely to develop a more favorable attitude toward the product than those exposed to NWOMnc and Adcc.

Methodology

The study employed a 2×2 experimental design to test the interaction effects between NWOM (Negative Word-of-Mouth) and advertising formats on consumer attitudes. This design enabled a focused examination of the effects of comparative and non-comparative NWOM (NWOMcc and NWOMnc) alongside attribute-based comparative and non-comparative advertising (Adcc and Adnc). While the design was well-suited for testing the hypothesized interaction effects, further elaboration on the sample and manipulation checks can enhance the robustness and generalizability of the findings.

Sample size and sampling method

The sample comprised 100 university students, aged 17–26 years ($M=20$, $SD=1.86$), with equal representation of male and female participants. This convenience sample was drawn from a university setting, aligning with the study's focus on low-involvement products like hands-free devices, which are commonly used among students. While this demographic provided a controlled environment for testing the hypotheses, the findings may have limited generalizability to broader populations. The reliance on university students means the results predominantly reflect the attitudes and behaviors of young, educated individuals who are frequent users of digital platforms.

Manipulation checks overview

To ensure the internal validity of the experimental conditions, comprehensive manipulation checks were conducted during the pretest phase. These checks aimed to confirm that participants accurately perceived the comparative and non-comparative formats of NWOM and advertising stimuli as intended.

NWOM manipulation check

Comparative and non-comparative advertising and NWOM formats were chosen for this study due to their differing cognitive effects. Comparative formats enhance diagnosticity by directly contrasting brands, whereas non-comparative formats present product information in isolation, allowing consumers to form independent judgments (Azemi et al., 2020; Nam et al., 2020). Given that NWOM itself can be comparative or non-comparative, this study examines how different advertising formats interact with these NWOM structures to influence consumer attitudes. Therefore, the participants were asked to evaluate the negativity and format of the NWOM content using a Likert scale and direct questions. The negativity of NWOM was assessed on a 15-point Likert scale ranging from -7 (very negative) to $+7$ (very positive), with results confirming the recognition of negative information across NWOMcc and NWOMnc conditions. Additionally, participants were asked whether the NWOM statements compared two brands (yes/no), confirming their understanding of the comparative or non-comparative nature of the content.

Advertising manipulation check

Attribute-based advertising was chosen for its alignment with cognitive information processing. Unlike emotional appeals, attribute-based formats provide rational, detailed product information, which allows consumers to engage in deeper cognitive evaluation—particularly relevant in countering NWOM that also carries cognitive content.

Attribute-based advertising focuses on providing detailed information about product characteristics, such as specifications, materials, and technical performance, rather than promoting abstract benefits like convenience or status (Ngo et al., 2024; Wiese et al., 2020). This study employs attribute-based advertising as it aligns with the cognitive nature of NWOM, allowing consumers to assess product details objectively. The choice of this format is further supported by prior research indicating that when consumers encounter NWOM, they seek factual counterarguments to reassess their perceptions (Birim et al., 2024; Ribeiro & Kalro, 2023). Moreover, Cognitive advertising refers to advertising messages that prioritize rational, analytical, and attribute-based information processing rather than emotional appeals. Grounded in cognitive psychology, this approach engages consumers' central processing routes, as proposed in the Elaboration Likelihood Model (Petty & Cacioppo, 1986). By presenting detailed product attributes, cognitive advertising facilitates deeper consumer engagement, particularly when motivation and ability to process information are high. Therefore, the participants were asked whether the ads presented a comparison between two brands (yes/no). To validate the cognitive nature of the advertising content, participants rated items such as whether the advertisement prompted analytical thinking or emphasized specific product attributes. Responses were collected on a 7-point Likert scale, ensuring that both Adcc and Adnc formats were perceived as intended.

Overall cognitive information manipulation

Both NWOM and advertising stimuli were evaluated for their cognitive content using a series of items adapted from established scales. These items assessed whether the stimuli encouraged participants to analyze product differences, consider arguments for or against the brand, and engage in critical thinking. The findings indicated that the manipulations were successful, with participants correctly distinguishing between the experimental conditions.

These manipulation checks strengthen the internal validity of the study, ensuring that the observed effects can be attributed to the intended variations in NWOM and advertising formats rather than misinterpretations of the stimuli. By incorporating robust validation procedures, this study provides a solid foundation for examining the interaction effects of NWOM and advertising on consumer attitudes. The variables of this study and their manipulation test are discussed in detail.

NWOMcc and NWOMnc

NWOMcc and NWOMnc were the independent variables. In this experiment, the negative information in the 'NWOMcc and NWOMnc' consumer review was checked by asking respondents to rate, 'How negative/positive the consumer review about the Ambits is?' on Likert scale of 15-point, ranging from -7 to +7 (where -7 is very negative and +7 is very positive) (Boonthanom, 2004). After that, the manipulation of comparative information in the NWOMcc was checked by simply asking the participants 'Do you agree the consumer review compares Ambits with Dacom? Yes/No'. The cognitive information manipulation in 'NWOMcc and NWOMnc' was checked on a 7 point Likert scale from '1 - Not at all' to '7 - Very much' adopted from (Chaudhuri & Buck, 1995). The following items were included: (1) Did the consumer review prompt you to consider the actual differences between Ambits and Dacom? (2) Did the consumer review lead you to contemplate the reasons supporting Ambits' superiority? (3) Did consumer reviews encourage you to evaluate the advantages and disadvantages of Ambits? (4) Did the consumer review stimulate analytical thinking rather than evoke emotional responses? (5) Did the consumer reviews reflect on arguments for or against the use of Ambits?

Adcc and Adnc

Adcc and Adnc are the independent variables. In this experiment, the manipulation of comparative information in Adcc was checked by simply asking the participants 'Do you agree the advertisement compares Ambits with Dacom? Yes/No'. The cognitive information manipulation in Adcc and Adnc was checked on a 7 point Likert scale from '1 - Not at all' to '7 - Very much' adopted from (Chaudhuri & Buck, 1995). The following items were included: (1) Did the advertisement prompt you to consider the actual differences between Ambits and Dacom? (2) Did the advertisement lead you to reflect on the reasons supporting Ambits' superiority? (3) Did the advertisement encourage you to evaluate the benefits and drawbacks of Ambits? (4) Did the advertisement stimulate critical thinking rather than an emotional response? (5) Did the advertisement contemplate arguments for or against the use of Ambits?

Attitude – 'attitude toward the product'

In this experiment, the sole dependent variable was 'Attitude toward the product.' This was assessed using a four-item, 7-point Likert scale, adapted from MacKenzie and Lutz (1989). The items used to measure the construct included 'What is your overall evaluation of Ambits?' with responses ranging from (1) Dislike to Like, (2) Negative to Positive, (3) Useless to Useful, and (4) Unfavorable to Favorable.

Research design and overview

An experimental approach was used to conduct this study. A two-factor design was employed in this experiment. The treatment conditions set for the study as elaborated in Figure 2 presents cognitive 'NWOM' (comprising NWOMcc and NWOMnc) and cognitive advertising (Adcc and Adnc). A '2 (cognitive NWOM) × 2 (cognitive advertising) mixed design; was employed to test hypotheses H1 and H2.

Attribute-Based Cognitive Advertising		
Negative Word of Mouth	ADcc	ADnc
	NWOMcc	NWOMcc/ADnc
	NWOMnc	NWOMnc/ADnc

Figure 2. Research design.

To execute this experiment, a hands-free brand was chosen as the product category. Two criteria were used to select this category for the study. First, it was generally observed that hands-free is a commonly used product among university students. The transition from traditional classroom learning to online education has significantly increased the use of personal listening devices such as hands-free sets, headphones, and earplugs, which was further supported by the findings of earlier research conducted on university students in Karachi (Asghar et al., 2022; Zia et al., 2014), which revealed that most university students use hands-free devices within the university and connect them to their mobile phones and laptops. Second, the hands-free is assumed to be a low involvement product; therefore, attribute-based cognitive advertising can be applied more effectively (Asghar et al., 2022; Eslami & Ghasemaghaei, 2018).

As our focus is on cognitive information content, choosing this product as a stimulus is appropriate because it does not stimulate emotional connectivity. Two fake brands, Ambits and Dacom were created, because taking existing brands into account for this study can lead to participants' bias. Ambits is assumed to be the competing brand and Dacom is assumed to be the leading brand. Two online shopping websites were created for Ambits. Each website had two pages; one for NWOM content, and another for advertising content.

On the first website, the NWOMcc and Adcc were placed separately on each page, and on the second website, the NWOMnc and Adnc were placed separately on each page.

The model developed for this experimental research was tested by distributing the respondents into four experimental designs (see [Supplementary Appendices 1–4](#)). The sets of stimuli for these four experimental designs were 'negative word of mouth comparative content and advertising comparative content (NWOMcc and Adcc)', 'negative word of mouth comparative content and advertising non-comparative content (NWOMcc and Adnc)', 'negative word of mouth non-comparative content and advertising comparative content (NWOMnc and Adcc)', and 'negative word of mouth non-comparative content and advertising non-comparative content (NWOMnc and Adnc)'. In the NWOMcc and Adcc experiment designs Ambits Hands-free was directly compared with Dacom Handsfree. However, the NWOMnc and Adnc experimental designs did not provide any information related to the Dacom Hands-free.

University students participated in the experiment and were randomly selected for one of four experimental designs. Before proceeding with the experiment, the objective of the experiment was explained to students. The objective of this experiment was explained to students. It was communicated to students that this study had been conducted to study consumer responses to a particular product or brand attribute. They were then exposed to their assigned set of cognitive NWOM content and attribute based cognitive advertisement via the web pages created earlier for this experiment. They were first asked to visit the web link for cognitive NWOM content and answer a few questions and then they were asked to visit the second web link for attribute-based cognitive advertisement content and answer the remaining questions.

For both the pretest and main study, the survey was conducted online using Google Forms. The form further contained the link of stimuli created for this research (See [Supplementary Appendices 1–4](#)). The questionnaire was composed of three main sections: Negative WOM Communication (Cognitive Information Content: Comparative/Non-Comparative Information Format), Attribute Based Advertising

Communication (Cognitive Information Content: Comparative/Non-Comparative Information Format), and Attitude toward the Product. The questionnaire was administered to students via the social media application.

This study was approved by the Research Ethics Review Board at the Faculty of Social Sciences, Bahria University, in December 2022. The purpose of the questionnaire was to examine the influence of Negative WOM Communication and Attribute Based Advertising Communication on Attitude toward the Product. Further the study also examined the moderating role of Attribute Based Advertising Communication.

Written informed consent was obtained from all participants prior to their involvement in the study. The questionnaire included a statement informing participants about the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without consequences. By proceeding with the questionnaire, participants provided their written consent to take part in the study.

To ensure the privacy and confidentiality of the participants, no personally identifiable information was collected during the survey. The data were anonymized and stored securely, with access restricted to the researchers directly involved in the study. The study adhered to the ethical guidelines and principles set forth by Bahria University and the Faculty of Social Sciences, ensuring the protection of participant privacy and confidentiality throughout the research process.

The study was approved by the Research Ethics Review Board for Research Involving Human Participants by Bahria University.

Pretests for stimuli development and selecting product attributes

For selecting the relevant product attributes of the stimuli, a questionnaire was developed to assess the chosen seven attributes for Ambits. These seven attributes were chosen based on consumers' reviews of hands-free brands on Amazon and Ali Express online shopping websites, which included soft ear cushion, flexible cord, volume adjustment option, multiple pin option, favorite colors, long wires and light weight. Initial website and advertisement stimuli were also developed, in which all these attributes were incorporated. In the first pretest, 50 students participated to determine whether the attributes chosen for the stimuli creation were important for consumers. The significance of the product attributes mentioned in the word of mouth statement and the advertisement was checked on a 7 – point Likert scale ranging from '1 - unimportant to 7 - important'.

Before completing the questionnaire, the participants were first asked to visit the web link to see NWOM and an advertisement placed on the website, they were then asked to suppose that they were going to buy a hands-free brand. They were further told to rate each attribute in terms of its importance to them. The descriptive analysis for Pretest 1 is summarized in Table 1.

As shown in Table 1, the results suggest that three product attributes are more significant: Flexible Cord (Mean, 5.80), Soft Ear Cushion (Mean, 5.56) and Volume Adjustment Option (Mean, 5.44). The information present in the stimuli content was revised based on these results and four other attributes were dropped to make the stimuli more precise for the respondents to answer.

NWOM comparative and non-comparative cognitive information

Fifty-two students participated in the second pretest. The objective of this pretest was to check the negativity of the message, cognitive information, and the comparative and non-comparative formats of NWOM content used in this experiment. The participants were divided into two groups. One group was exposed to NWOMcc and the other group was exposed to NWOMnc. As in the first pretest, the

Table 1. Pretests 1 for stimuli development and selecting product attributes.

Product attributes	Descriptive analysis			
	Mean	Standard deviation	Skewness	Kurtosis
Soft ear cushion	5.56	1.75	−1.14	.37
Flexible cord	5.80	1.60	−1.28	.90
Volume adjustment option	5.44	1.94	−0.94	−0.27
Multiple pin option	4.45	1.92	−0.37	−1.19
Favorite color	3.47	1.80	.31	−0.76
Long wires	4.71	1.71	−0.39	−0.52
Light weight	4.82	1.75	−0.73	−0.27

participants were provided with the web link and asked to follow the same instructions for filling out the questionnaire as stated in Pretest 1. Participants in both groups were asked to recognize the negative information in the content placed on the website on a 15-point Likert scale with scores ranging from -7 to $+7$ (where -7 is very negative and $+7$ is very positive) (Boonthanom, 2004). The participants who were exposed to NWOMcc recognized negative information ($M=-2.26$, $SD=4.13$), and the participants of another group who viewed NWOMnc content recognized the negative information ($M=-2.30$, $SD=3.41$). 88% of participants, who were exposed to the NWOMcc recognized the comparative information format. Cognitive information manipulation in both NWOMcc and NWOMnc was checked by running a one-sample t-test, but before that normal distribution of the data was checked. The findings are summarized in Table 2.

As shown in Table 2, the data were normally distributed. A one-sample t-test was run in SPSS to check the manipulation of cognitive information content in negative word of mouth communication (Comparative Content) and negative word of mouth communication (Non-Comparative Content). The analysis using the one-sample t-test was conducted to validate the manipulation of cognitive information content in the negative word of mouth (NWOM) stimuli, specifically comparing the comparative content (NWOMcc) and non-comparative content (NWOMnc). The purpose was to determine whether participants' perceptions of the cognitive information in the stimuli significantly differed, ensuring that the stimuli accurately reflected the intended manipulation for each condition.

The results indicated a statistically significant difference between the mean scores of cognitive information content in both conditions. For NWOMcc, the mean score ($M=4.50$, $SD=1.03$) was significantly different from the baseline value, as indicated by the t-test result ($t(4) = 2.50$, $p=0.01$). Similarly, for NWOMnc, the mean score ($M=4.33$, $SD=0.411$) also showed a significant difference ($t(4) = 4.16$, $p=0.00$). These findings confirm that the cognitive information content in both comparative and non-comparative formats was effectively manipulated, providing validity to the stimuli used in the subsequent main study.

This analysis ensured that the participants perceived the cognitive information as intended, laying a robust foundation for testing the hypotheses in the main study. The findings are summarized in Table 3.

Advertisement comparative and non-comparative cognitive information

Fifty-two students participated in the third pretest. The objective of this pretest was to check the cognitive information and the comparative and non-comparative format of Ad content used in this experiment. The participants were divided into two groups. One group was exposed to Adcc and the Adnc content was shown to the other group. As in the second pretest, the participants were provided with the web link and asked to follow the same instructions for filling out the form as stated in pretests 1 and 2. All participants exposed to Adcc recognized the comparative information format. Cognitive information manipulation in both Adcc and Adnc was checked by running a one-sample t-test, but before that, normal distribution of the data was checked. The findings are summarized in Table 4.

As shown in Table 4, the data followed a normal distribution. A one-sample t-test was conducted using SPSS to evaluate the manipulation of 'cognitive information content' in Adcc and Adnc groups. This analysis using the one-sample t-test was conducted to validate the manipulation of cognitive

Table 2. NWOM comparative and non-comparative cognitive information.

Information type	Descriptive analysis			
	Mean	Standard deviation	Skewness	Kurtosis
NWOMcc	4.50	1.03	-1.5	-0.83
NWOMnc	4.33	.41	-4.3	-4.1

Table 3. NWOM comparative and non-comparative cognitive information.

Information type	One sample t-test		
	Mean	Standard deviation	p value
Cognitive Information Manipulation in NWOMcc	4.50	1.03	0.01
Cognitive Information Manipulation in NWOMnc	4.33	.41	0.00

$p < 0.05$.

information content in the attribute-based advertising stimuli, specifically comparing the comparative advertising content (Adcc) and non-comparative advertising content (Adnc). The objective was to ensure that the participants' perceptions of cognitive information in the advertising stimuli significantly differed, confirming the effectiveness of the intended manipulations.

The results indicated a statistically significant difference between the mean scores of cognitive information content for both conditions. For Adcc, the mean score ($M=5.77$, $SD=0.41$) significantly differed from the baseline, as shown by the t -test result ($t(4) = 21.73$, $p=0.00$). Similarly, for Adnc, the mean score ($M=4.96$, $SD=0.88$) also demonstrated a significant difference ($t(4) = 5.55$, $p=0.00$). These findings confirm that the participants recognized the cognitive information content in both comparative and non-comparative formats as intended.

This analysis validates that the advertising stimuli were appropriately manipulated to reflect the comparative and non-comparative conditions, ensuring that the experimental setup was robust and suitable for testing the hypotheses in the main study. The findings are summarized in Table 5.

The use of the One Sample t -test in this study was limited to the pretest phase, specifically to validate the manipulation of cognitive information within the stimuli. The test was conducted to determine whether the mean responses for comparative and non-comparative cognitive information formats significantly differed, thereby confirming the effectiveness of the stimuli. This step was essential to ensure that the experimental conditions were accurately operationalized and aligned with the study's objectives before proceeding to the main experiment.

It is important to clarify that the One Sample t -test was not utilized in the primary analysis of the main study. Instead, Two-Way ANOVA was employed to examine the hypothesized relationships and test the interaction effects of negative word-of-mouth communication and attribute-based cognitive advertising on consumer attitudes. By distinguishing the role of the One Sample t -test as preparatory and explicitly aligning it with the pretest phase, this study adheres to methodological rigor and ensures the validity of the stimuli and subsequent findings.

Main study

Experimental procedure

In the final study, 100 students were randomly selected to participate. During the pretests, it was observed that students were reluctant to read the stimuli before answering questions. Therefore, before participating in the final study, the students were told that any four participants would be randomly selected and awarded a small gift once the data collection was completed. Participants were divided into four equal groups, and each group was exposed to the assigned condition. The experimental conditions included: NWOMcc combined with Adcc, NWOMcc combined with Adnc, NWOMnc combined with Adcc, and NWOMnc combined with Adnc. The responses used in the main study were different from the pretests. The data were collected simultaneously and used mutually for each hypothesis. The participants were aged between 17 and 26 years ($M=20$, $SD=1.86$). 50% of them were female ($N=50$) and 50% were male ($N=50$).

Table 4. Advertisement comparative and non-comparative cognitive information.

Information type	Descriptive analysis			
	Mean	Standard deviation	Skewness	Kurtosis
Adcc	5.77	.41	−0.32	−0.07
Adnc	4.96	.88	−0.21	−0.88

Table 5. Advertisement comparative and non-comparative cognitive information.

Information type	One sample t -test		
	Mean	Standard deviation	p value
Cognitive Information Manipulation in Adcc	5.77	.41	0.00
Cognitive Information Manipulation in Adnc	4.96	.88	0.00

$p < 0.05$.

Table 6. Results and discussion.

Statistical result of hypothesis No. 1			
Dependent variable	NWOMcc	NWOMnc	<i>p</i> value
Attitude toward the product	3.39(1.58)	3.61(1.76)	NS

Table 7. Results and discussion.

Statistical result of hypothesis No. 2					
Dependent variable	NWOMcc		NWOMnc		<i>p</i> value
	adcc	Adnc	adcc	Adnc	
Attitude toward the product	3.11(1.71)	3.68(1.42)	2.15(0.91)	5.07(1.02)	0.00

The reliability analysis of attitude toward the product was checked by running Cronbach's alpha test in SPSS. As mentioned earlier, the scale included four items: 'Dislike to Like,' 'Very Negative to Very Positive,' 'Useless to Useful,' and 'Very Favorable to Very Unfavorable' (MacKenzie & Lutz, 1989), assessed using a 7-point Likert scale. Cronbach's alpha for the 'Attitude toward the Product' measure was 0.90. The manipulation of comparative information was checked in both 'NWOMcc and Adcc.' 100% of the participants who were exposed to the 'NWOMcc' statement acknowledged the comparative information format. Additionally, all participants who viewed the 'Adcc' placed on the website identified the comparative information format.

Results

The hypotheses were tested using a two-way ANOVA in SPSS, with 'Attitude toward the Product' as the dependent variable. Hypothesis 1 posited that 'NWOMcc would result in a more negative attitude toward the product compared to NWOMnc.' However, the results showed no significant difference between the NWOMcc and NWOMnc regarding attitudes toward the product ($M_{NWOMcc} = 3.39$, $M_{NWOMnc} = 3.61$, $F = 0.67$, NS). Therefore, Hypothesis 1 is not supported. The findings are summarized in Table 6.

Hypothesis 2 proposed that 'attribute-based cognitive advertising moderates the impact of NWOM on consumer attitudes toward the product. Specifically, individuals exposed to NWOMcc and Adcc are expected to have more favorable attitude toward the product than those exposed to NWOMcc and Adnc. Additionally, those exposed to NWOMnc and Adnc should have a more favorable attitude than those exposed to NWOMnc and Adcc.' The results of the experiment indicated that attribute-based cognitive advertising moderated the effects of negative word-of-mouth communication on attitudes toward the product ($p = 0.00$). However, the observed interaction effect did not align with our predictions. Therefore, Hypothesis 2 is partially supported. The findings are summarized in Table 7.

Contrary to the anticipated more favorable interaction effect of 'NWOMcc and Adcc on Attitude toward the Product,' the results showed that participants exposed to 'NWOMcc and Adcc' actually developed a less favorable attitude toward the product compared to those exposed to 'NWOMcc and Adnc.' Conversely, individuals exposed to 'NWOMnc and Adnc' had a more favorable attitude toward the product than those exposed to 'NWOMnc and Adcc.'

Discussion

In today's digital environment, product information is becoming increasingly accessible to consumers through word-of-mouth communication and advertising. Hence, this research practically contributes to study the impact of 'NWOMcc and NWOMnc' on 'Attitude toward the Product,' and the moderating role of 'Attribute-based Cognitive Advertising.'

The findings provide critical insights into the role of comparative and non-comparative formats in shaping consumer attitudes. While Hypothesis 1 posited that NWOMcc would result in a more negative attitude compared to NWOMnc, the results indicate no significant difference. This suggests that

consumers may not necessarily perceive comparative NWOM as more damaging than non-comparative NWOM. One possible explanation is that consumers focus on the negative tone rather than the comparative structure of NWOM when forming attitudes toward a product.

Hypothesis 2, which examined the moderating role of attribute-based advertising, produced an unexpected interaction effect. While it was anticipated that NWOMcc combined with Adcc would mitigate negativity, the results indicated that NWOMcc paired with Adnc led to a more favorable attitude. This suggests that comparative advertising might reinforce the negativity of NWOMcc rather than counteracting it. On the other hand, NWOMnc combined with Adnc resulted in the most favorable attitudes, implying that when NWOM is less confrontational, a non-comparative advertising response is more effective in maintaining consumer trust.

These findings highlight the complexity of how comparative and non-comparative content interact in shaping consumer attitudes. While comparative NWOM was expected to be more damaging and comparative advertising more effective in countering NWOM, the results suggest that message framing plays a crucial role. The unexpected effectiveness of Adnc over Adcc indicates that non-comparative advertising may be perceived as more neutral and credible, whereas comparative advertising could reinforce NWOM's negative framing. This suggests that the mere presence of advertising provides an alternative narrative that helps mitigate NWOM's impact, regardless of format. Future research should explore additional cognitive mechanisms, such as skepticism and perceived credibility, to better understand how consumers process NWOM and advertising in different formats.

The research model for this study was derived on the basis of 'Elaboration Likelihood Model' (Petty & Cacioppo, 1984; Petty & Cacioppo, 1986) and the 'Accessibility-diagnosticity Model' (Lynch & Feldman, 1988). The results do not clearly support the assumption that 'cognitive information presented in a comparative format is more diagnostic than information presented in a non-comparative format.' Consequently, based on our findings, 'NWOMcc may not result in a more negative attitude toward the product compared to NWOMnc.' Similarly, contrary to the hypothesized interaction effect, Adcc may not lead to a more positive attitude toward the product than Adnc. Many other researchers have recommended that in order to form an attitude, consumers integrate incoherent information coming from various sources and consider each part of the information (Lautenbach & Heyder, 2019; Ma et al., 2022; Xiang et al., 2019). The results of the moderating effects of attribute-based advertising suggest that, when NWOMcc and NWOMnc occur, Adnc should be incorporated to minimize damage.

The Accessibility-Diagnosticity Model (ADM) and the Elaboration Likelihood Model (ELM) were central to the theoretical framework of this study, providing insights into how consumers process negative information (NWOM) and respond to advertising. The ADM posits that the likelihood of information influencing judgment depends on its accessibility and diagnosticity. Diagnostic information is that which consumers perceive as relevant and useful in decision-making (Haq et al., 2024). In this study, it was hypothesized that comparative NWOM (NWOMcc) would be more diagnostic than non-comparative NWOM (NWOMnc), given that direct comparisons provide detailed points of differentiation. Similarly, the ELM highlights that when consumers are highly involved, they are more likely to process information through the central route, which involves deep cognitive engagement. This suggested that cognitive, attribute-based comparative advertising (Adcc) would lead to more favorable attitudes under high-involvement conditions compared to non-comparative advertising (Adnc). Research has consistently highlighted the role of cognitive, attribute-based comparative advertising (Adcc) in shaping consumer attitudes, particularly under high-involvement conditions. Comparative advertising is known to prompt analytical processing by directly contrasting product attributes with competing brands, which enhances message elaboration and diagnosticity (Ghiassaleh et al., 2024; Yang et al., 2024).

However, the results challenged these theoretical expectations. NWOMcc did not produce significantly more negative attitudes compared to NWOMnc, and Adcc did not consistently mitigate NWOM's effects more effectively than Adnc. These findings emphasize the need to explore additional cognitive mechanisms that may explain the observed outcomes. One such mechanism is negativity bias, which suggests that consumers are more likely to focus on and be influenced by negative information than positive information, regardless of its presentation format (Grassini & Koivisto, 2024). In the case of NWOM, the negativity of the content may overshadow its comparative or non-comparative nature, leading to similarly negative consumer attitudes in both conditions. This aligns with findings of previous research, that

negative information often has a disproportionate impact on judgment, making it challenging for advertising to counteract its effects fully (Li et al., 2024; Watson et al., 2024).

Additionally, the context of attribute-based advertising and the **cognitive load** it imposes must be considered. Comparative advertising requires consumers to process multiple data points simultaneously to evaluate the differences between brands (Zhu et al., 2023). This additional cognitive demand may lead to cognitive overload, particularly when paired with the inherently attention-grabbing nature of negative information (Tarafdar et al., 2024). In such scenarios, consumers might default to heuristic processing, where they rely on general impressions of negativity rather than deeply analyzing the comparative claims. This cognitive overload could dilute the expected diagnostic advantage of comparative formats, explaining why Adcc did not perform as hypothesized.

Furthermore, the cultural context of this study provides critical insights. In Pakistan, where comparative advertising is less prevalent and often perceived as unethical, consumers may approach such advertisements with skepticism or distrust (Alam et al., 2024). This cultural lens likely influenced participants' processing of Adcc, rendering it less effective at mitigating NWOM's effects. Previous studies have suggested that cultural norms significantly shape consumer responses to advertising, and formats that deviate from these norms may face resistance (Rahman et al., 2023; Vidani et al., 2023). Consequently, the expected diagnosticity and central route processing benefits of comparative advertising may not have been realized in this context.

In conclusion, while ADM and ELM provide a robust foundation for understanding consumer information processing, their application in this study highlights the interplay of additional cognitive mechanisms and cultural factors. The findings suggest that the diagnosticity of comparative formats may be neutralized by the pervasive influence of negativity bias, the potential for cognitive overload, and culturally driven skepticism. Future research could explore these dynamics further by incorporating experimental conditions that manipulate cognitive load, test varying degrees of negativity, and account for cross-cultural differences in advertising effectiveness. These extensions would deepen our understanding of the theoretical and practical implications of NWOM and advertising strategies.

Conclusion

This study significantly contributes to the understanding of how Negative Word-of-Mouth (NWOM) and advertising formats interact to shape consumer attitudes, offering new insights into the cognitive and cultural factors influencing these dynamics. By integrating the Elaboration Likelihood Model (ELM) and the Accessibility-Diagnosticity Model (ADM), this research provides a robust theoretical framework for analyzing how consumers process NWOM in comparative and non-comparative formats and respond to advertising under varying conditions of cognitive involvement.

The findings highlight the critical role of negativity bias, which underscores the disproportionate weight consumers place on negative information, regardless of its format. This bias can override the diagnostic advantage of comparative NWOM (NWOMcc), resulting in similar effects on consumer attitudes compared to non-comparative NWOM (NWOMnc). Additionally, the results underscore the impact of cognitive overload, suggesting that the comparative nature of NWOMcc may impose additional cognitive demands, leading to superficial information processing rather than detailed evaluations. These mechanisms challenge the assumption that comparative formats are inherently more diagnostic, emphasizing the importance of considering cognitive constraints in consumer information processing.

From a cultural perspective, this study contributes to the limited body of knowledge on how advertising formats function in non-Western contexts like Pakistan. Here, where comparative advertising is less prevalent and often viewed with skepticism, its effectiveness in mitigating NWOM is diminished. The findings suggest that cultural norms and consumer familiarity with advertising formats play a critical role in shaping responses, highlighting the need for culturally sensitive marketing strategies.

Methodologically, this research fills a significant gap by employing a rigorous quasi-experimental design with carefully validated stimuli to test the interactive effects of NWOM and advertising. The use of controlled experimental conditions ensures the robustness of the findings while providing a basis for future studies to build upon. The inclusion of manipulation checks further strengthens the internal validity, confirming that the observed effects are attributable to the intended experimental manipulations.

Practically, the study offers actionable insights for marketers seeking to mitigate the adverse effects of NWOM in digital settings. It emphasizes the value of non-comparative advertising formats, particularly in cultural contexts where comparative approaches may be perceived as untrustworthy or inappropriate. Marketers are advised to focus on reducing the salience of negativity in NWOM and leveraging advertising strategies that are aligned with consumer expectations and cultural norms.

In addressing these gaps, this research advances the academic discourse on NWOM and advertising, providing an in-depth understanding of their interplay. It establishes that while ADM and ELM are valuable frameworks, their application must be contextualized within the broader cognitive and cultural landscape. Furthermore, the findings pave the way for future research to explore additional variables such as consumer skepticism, brand loyalty, and product involvement to enrich the theoretical and practical implications.

In conclusion, this study bridges critical gaps in understanding the interactive effects of NWOM and advertising formats on consumer attitudes. By doing so, it not only contributes to the theoretical refinement of ADM and ELM but also provides a foundation for culturally adaptive and cognitively informed marketing practices in the increasingly dynamic digital landscape.

Limitations

This study is limited by its sample size and demographic focus on university students, which may not fully capture broader consumer responses. Additionally, while the study focused on attribute-based advertising, it did not explore how other advertising types might interact with NWOM. Further, only experimental research methods have been used in this research. These limitations present opportunities for future research to test the findings across diverse populations, advertising strategies and research methodologies.

The sample comprised a convenience sample of university students aged 17–26 years. While this demographic is relevant for studying low-involvement products like hands-free devices, it limits the generalizability of the findings to broader populations, such as older adults, professionals, or individuals from diverse educational and socioeconomic backgrounds. Second, the study focused exclusively on a single product category—a low-involvement product—which restricts the applicability of the findings to other product types. High-involvement or emotionally driven products, for example, may elicit significantly different consumer behaviors and responses to NWOM and advertising formats.

Another limitation lies in the controlled nature of the experimental environment. The use of carefully designed online stimuli ensured internal validity but does not fully replicate the complexities of real-world contexts. In practical settings, consumers encounter NWOM and advertising in more dynamic and varied scenarios, which can influence their attitudes in ways that controlled experiments may not capture. Furthermore, the study utilized a single NWOM message for each condition. In reality, consumers are often exposed to multiple and potentially conflicting NWOM messages from various sources, which may interact differently to influence attitudes. The reliance on a single message limits the ecological validity of the findings.

Further, the cultural context in which the study was conducted also presents a limitation. The research was carried out in Pakistan, where comparative advertising is less common and often viewed as unethical. This cultural backdrop likely influenced participants' responses, which could differ significantly in regions where comparative advertising is more prevalent or culturally accepted. While this study discusses the potential influence of cultural context, it does not directly measure cultural attitudes toward comparative advertising. Additionally, the study focused solely on cognitive information in NWOM and advertising, excluding affective elements such as emotional appeals or the tone of NWOM. This exclusion limits the scope of the findings, as emotional factors play a significant role in consumer decision-making processes.

One more limitation of the study relates to the variation in message length across the experimental conditions. Comparative conditions (cc), by design, contained more detailed content to enable brand comparisons, whereas non-comparative conditions (nc) were more concise. This discrepancy in word count may have imposed a higher cognitive load in the cc conditions, potentially influencing information processing and attitude formation. While pretests confirmed the effectiveness of the stimulus manipulation, future studies should aim to match word count across conditions or statistically control for message length to isolate the specific effects of comparative versus non-comparative framing.

Future research

Future research should address the limitations of this study to enhance the generalizability and depth of understanding in this area. Expanding the demographic diversity of participants is essential. Future studies should include individuals from various age groups, educational levels, cultural backgrounds, and cross-cultural comparisons to ensure a broader representation of consumer behavior. They should further control the message length to exclude the effects of comparative versus non-comparative information formats. Additionally, exploring the interactive effects of NWOM and advertising across different product categories, including high-involvement or emotionally charged products, could provide valuable insights into how product characteristics influence consumer attitudes.

Real-world settings should also be incorporated into future research designs. Simulating dynamic and realistic scenarios where consumers are exposed to multiple and inconsistent NWOM messages from varied sources could better capture the complexity of consumer decision-making. This approach would allow researchers to examine how consumers process and integrate conflicting information to form attitudes. Similarly, cross-cultural research could provide a comparative perspective on how cultural norms influence responses to NWOM and advertising. For instance, studying contexts where comparative advertising is widely accepted versus regions where it is not could offer new insights into cultural differences in information processing.

Future studies should also include affective elements in their examination of NWOM and advertising. Emotional appeals in advertising and the tone of NWOM are critical aspects of consumer decision-making and should be integrated alongside cognitive content for a more holistic understanding. Incorporating variables such as brand loyalty, consumer skepticism, and product familiarity could further enrich the analysis by capturing individual differences in consumer responses. Finally, conducting longitudinal studies to assess the long-term effects of NWOM and advertising on consumer attitudes and purchase behaviors would provide a deeper understanding of how these interactions evolve over time. These efforts will build upon the foundation established by this study and contribute to advancing both theoretical and practical insights in the field.

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Authors contributions

CRedit: **Safeena Yaseen**: Conceptualization, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Writing – original draft, Writing – review & editing; **Smith Boonchutima**: Formal analysis, Investigation, Resources, Supervision, Validation, Writing – review & editing; **Ibtesam Mazahir**: Data curation, Formal analysis, Methodology, Project administration, Resources, Software, Validation, Writing – review & editing.

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

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